LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

ALTERNATE FLOW SPECIFICATION

Problem Statement #24 — Smart Cities, Transport & Logistics

Public Bus Live Tracking & Crowding Estimator

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SECTION: ‘H’

Use Case: UC-03 — View Bus Information

Primary Actor: Commuter

Goal:
Allow the commuter to view the current location, estimated arrival time (ETA), and passenger crowding level of buses operating on a selected route.

Related Functional Requirements

- **FR-001:** The system shall ingest bus GPS coordinates every 5 seconds and recalculate arrival ETAs for all upcoming route stops.
- **FR-002:** The system shall estimate the passenger crowding level of each active bus using data received from ticketing sensors.
- **FR-003:** The system shall allow commuters to view the current location, estimated arrival time, and crowding level of buses operating on a selected route.

These requirements are consistent with the functional requirements already documented in the submitted requirements analysis.

Alternate Flow A1 — GPS Data Unavailable

Trigger

The system does not receive valid or recent GPS coordinates from the selected bus.

Flow

1. The commuter selects a bus route or a specific bus.
2. The system attempts to retrieve the latest GPS coordinates.
3. The system detects that valid GPS data is unavailable.
4. The system cannot determine the current bus location reliably.
5. The system cannot calculate a reliable current ETA.
6. The system displays a message:

"Live bus location and ETA are temporarily unavailable."

7. If previously recorded information is available, the system displays the last known bus location and information.
8. The system continues monitoring for new GPS data.
9. When valid GPS data becomes available, the system updates the bus location.
10. The system recalculates the ETA for upcoming stops.
11. The updated information is displayed to the commuter.
12. The use case continues normally.

Expected Outcome

The commuter is informed about the temporary unavailability of live GPS information instead of receiving an incorrect bus location or ETA.

Alternate Flow A2 — High Crowding Detected

Trigger

The ticketing sensor data indicates that the selected bus has reached a high crowding level.

Flow

1. The system receives updated ticketing sensor data.
2. The system calculates the current passenger crowding level.
3. The system detects that the crowding level is above the configured threshold.
4. The system classifies the bus as highly crowded.
5. The system displays the high crowding status to the commuter.
6. The commuter can use the information when deciding whether to board the selected bus.
7. The system continues monitoring new ticketing sensor data.

Expected Outcome

The commuter receives updated crowding information and is made aware that the selected bus has a high passenger load.

Alternate Flow Summary

The alternate flows allow the system to handle conditions in which normal processing cannot be completed exactly as expected. GPS unavailability results in temporary unavailability of live location and ETA information, while high crowding results in the system displaying an appropriate crowding status.